

M Navaneeth Krishna

Senior Backend Engineer | Data Systems & Distributed Architecture | IIT Madras (BS Data Science)

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PROFESSIONAL SUMMARY

Backend Engineer with 4+ years building **data-intensive distributed systems** for fintech applications. Architected **Stripe-powered corporate card infrastructure** with real-time transaction processing, spending controls, and automated ledger reconciliation. Experience designing **real-time streaming pipelines**, multi-tenant data architectures, and high-throughput reporting engines. Strong foundation in **payment systems**, data integrity, and ACID compliance. Currently pursuing **BS in Data Science** from IIT Madras.

TECHNICAL SKILLS

Backend & Data: C#, Python, SQL, REST APIs, Entity Framework, Stripe API, Background Services, Data Processing

Databases: MSSQL, PostgreSQL, Redis (Caching), Firebase — Data Warehousing, Query Optimization, Indexing

Distributed Systems: Multi-Tenant Architecture, Azure Service Bus (Kafka-equivalent), Event-Driven Design, SignalR (Real-Time)

Cloud & DevOps: Azure (Service Bus), Docker, Docker Compose, CI/CD (GitHub Actions), Firebase, Containerized Deployments

Architecture: Domain-Driven Design, CQRS, Repository Pattern, Clean Architecture, Modular Monolith

Learning/Exploring: Apache Kafka, Apache Spark, Data Lake Architecture, AWS/GCP Services

EXPERIENCE

Senior Software Engineer L1

Jan 2025 - Present

GeekyWolf | Trivandrum, Kerala, India

Lead Backend Developer — Cloud Accounting Platform

Data Pipeline & Reporting Engine

- Built **financial data pipeline** processing transactions into real-time P&L, Balance Sheet, Trial Balance, and Cash Flow reports
- Designed **ETL workflows** for bulk data imports (CSV/Excel) — opening balances, contacts, historical transactions
- Architected **data warehousing layer** for historical financial analytics, aggregated reporting, and trend analysis
- Implemented **background job processors** for scheduled report generation, email ingestion, and data reconciliation
- Created **materialized views** and Redis-cached report snapshots for low-latency dashboard queries

Distributed Multi-Tenant Architecture

- Architected **tenant-isolated data stores** with per-tenant configurations, chart of accounts, and access controls
- Implemented **Role-Based Access Control (RBAC)** with Business → Users → Roles permission hierarchy
- Built **audit trail system** tracking all data mutations with user context, timestamps, and change history

Real-Time Streaming & Event Processing

- Implemented **Azure Service Bus** (Kafka-equivalent messaging) for async event processing across distributed services
- Integrated **SignalR** for real-time data streaming — live dashboard updates, notifications, collaborative alerts
- Built **email ingestion pipeline** using Brevo API with background listeners auto-processing incoming invoices
- Implemented **event-driven architecture** with domain events triggering downstream data updates

Stripe Payment Infrastructure (Core Feature)

- Architected end-to-end **Stripe Connect** integration for multi-tenant business payment processing
- Built complete **corporate card issuance system** — provision branded cards per business, assign to projects/employees
- Implemented **real-time spending controls**: set/update limits, freeze/unfreeze cards, block categories, enforce policies
- Designed **project-based card allocation** — each project gets dedicated card with isolated budget tracking
- Built **webhook ingestion pipeline** processing card transactions in real-time, auto-posting to double-entry ledger
- Implemented **transaction categorization** and merchant data enrichment for expense analytics
- Created card lifecycle management: issuance → activation → spending → reconciliation → deactivation

Data Integrity & Compliance

- Enforced **ACID transactions** on all financial mutations with optimistic concurrency control
- Double-entry ledger with automatic journal posting, debit/credit balancing, and period closing
- Soft deletes with full history retention for regulatory compliance and audit requirements

Infrastructure & Tools

- **Stack:** ASP.NET Core • MSSQL • Redis • Azure Service Bus • Docker • Firebase Auth • DDD + CQRS
- Containerized multi-service deployment with Docker Compose (API, Database, Cache, Workers)
- Built **MediateX** — in-process message passing library for request/response handlers and pipeline behaviors

Sr. Software Developer

Feb 2024 - Jan 2025

Ivory Technologies Pvt. Ltd. | Trivandrum, Kerala, India

- Developed enterprise applications with **offline-first data synchronization** and conflict resolution
- Implemented **local caching strategies** with background sync for intermittent connectivity
- Set up CI/CD pipelines for automated builds, testing, and deployment workflows

Software Developer

Feb 2023 - Jan 2024

Gmagica | Delhi, India

- Built applications consuming REST APIs with proper error handling, retry mechanisms, and data caching
- Collaborated with backend teams on API design and data contract specifications

Software Engineer

Aug 2022 - Mar 2023

Ykipp (Talrop) | Malappuram — Clean architecture, code reviews, Agile development

Junior Project Coordinator

Oct 2021 - Aug 2022

Tegain | Irinjalakuda — Project management, stakeholder communication, Agile sprints

RELEVANT EXPERTISE

Data Processing: Financial transaction pipelines • Data warehousing • Stripe webhook ingestion • Report engines • ETL • Batch processing

Payment Systems: Stripe Connect • Corporate card issuance • Real-time spending controls • Transaction reconciliation • PCI-aware design

Distributed Systems: Multi-tenant data isolation • Azure Service Bus (Kafka-equivalent) • Event-driven architecture • Background workers

Streaming: Message queues • Real-time data updates (SignalR) • Email ingestion pipelines • Live dashboard feeds

Data Integrity: ACID compliance • Audit trails • Soft deletes • Optimistic concurrency • Double-entry validation

Cloud/DevOps: Docker containerization • Multi-service orchestration • CI/CD automation • Firebase Cloud

EDUCATION

BS in Data Science and Applications — Indian Institute of Technology Madras

2025 - 2029